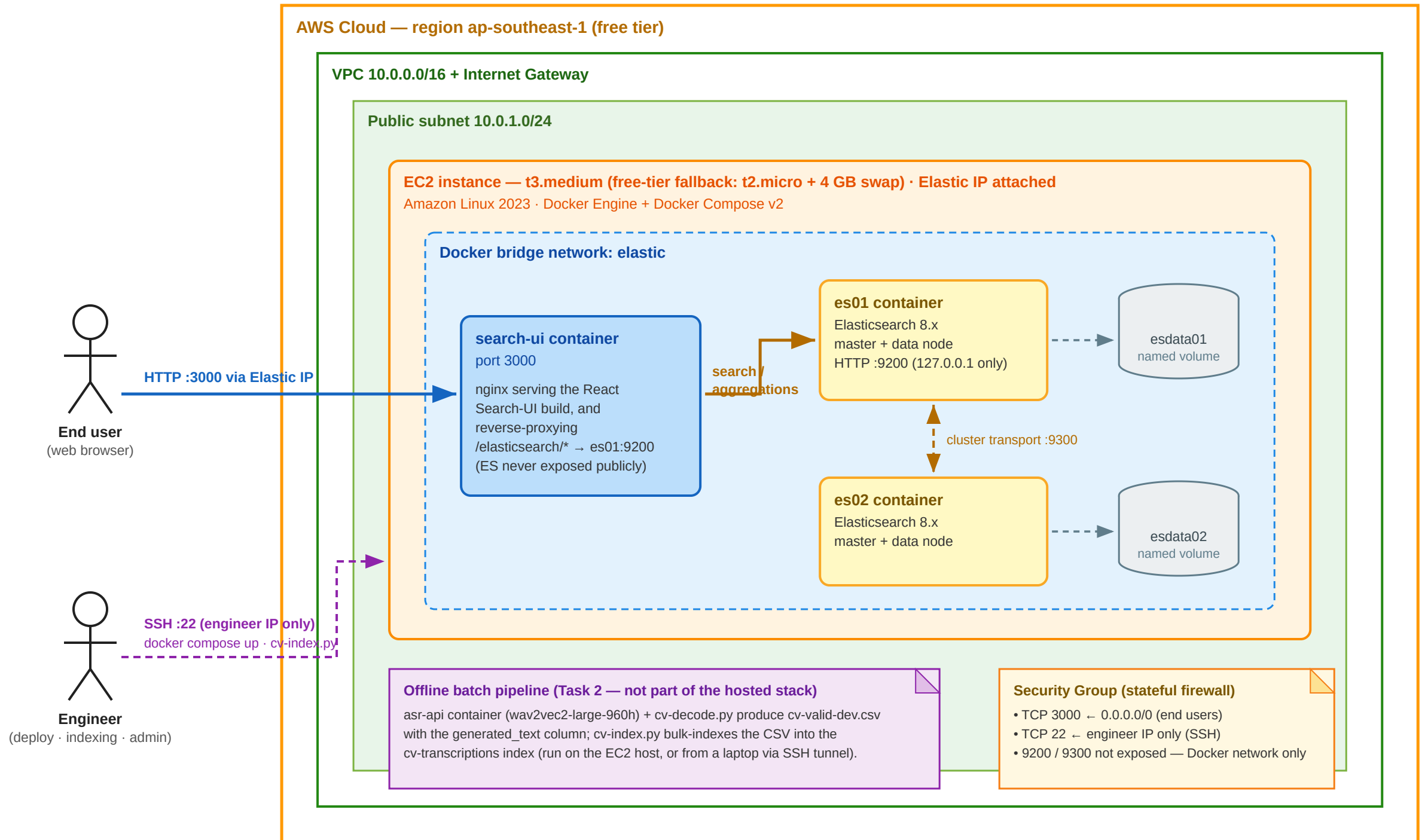


HTX xData — Search Service Deployment Architecture

AWS · self-managed Docker containers only (no managed services) · drawn with draw.io (source: design.drawio)



Design rationale

1. No managed services (per requirement): Elasticsearch and the frontend run as plain Docker containers on a single EC2 VM — no OpenSearch Service, no ECS/EKS, no ALB.
2. Only port 3000 (and admin SSH) is reachable from the internet; the browser talks to Elasticsearch exclusively through the nginx reverse proxy inside the search-ui container.
3. The 2-node ES cluster (replicas=1) survives the loss of one data node; named volumes persist the index across container restarts; Elastic IP keeps the public URL stable.
4. Free-tier note: a t2.micro (1 GB RAM) is possible with reduced ES heap (256 MB/node) + swap, but t3.medium is recommended for a responsive demo.